Agentic AI Bootcamp

Research Workflows, Teaching & Applications

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thinkingwithagents.github.io

Where We Left Off & Today's Plan

Session 1

- Agentic AI vs. chat AI
- Context window & planning
- `CLAUDE.md`, `PLAN.md`, `SKILL.md`
- Permissions & safety
- Apps: voice file, website, lecture builder spec

Emily: Research

1. **Code audit** — 51 files to 8, found a real bug
2. **Paper audit** — AI vs. 11 real referee reports
3. **New analysis** — Lee bounds, live

One project, three use cases.

Erkmen: Teaching

1. **Lecture Builder** — papers to slides
2. **Research Brainstorm** — stress-test live
3. **Web Scraper** — novel data in 10 min

Setup: Install Your Skill Toolkit

Anthropic skills (one-time)

1. **Add the marketplace:**

```
/plugin marketplace add anthropics/skills
```

2. **Install the document-skills bundle:**

```
/plugin install
  document-skills@anthropic-agent-skills
```

Gives you: `pdf`, `docx`, `pptx`, `xlsx`, `frontend-design`, `skill-creator`, `algorithmic-art`.

Third-party skills via `npx`

Full economist pack:

```
npx skills add thinkingwithagents/skills
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One skill at a time:

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Includes: `research-brainstorm`, `find-data`, `academic-beamer-deck`.

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! `/skill-creator` is **not** bundled with Claude Code by default — it ships inside `document-skills`. Install it before App 3.

Emily: Research Workflows

Code Auditing, Paper Auditing & Analysis

Application 1

Code Audit & Refactoring

Bangladesh Remote Education RCT

The Project

- Paper on remote education in Bangladesh
- Evaluates TV instruction, adaptive edtech, data subsidies, and teacher support during COVID-era school closures
- **Keeps getting rejected** — 8 submissions 😫
- One project, three AI use cases

Three use cases:

1. Code audit
Restructure 6 (😱) years of accumulated code into a clean replication package

2. Paper audit
Adversarial review — catch what referees would catch

3. New analysis
Referee says "do Lee bounds" — so we do it, live

Code Audit: Before

```
02_Do files/
01_master_5th June 2021.do
03_vargen_07sep.do
03_vargen.do
05_endline balance_eb.do
05_endline balance.do
09_attrition_8 July_e.do
09_attrition_8 July.do
...
Archive/ (18 files)
WB Submission Files/ (14 copies)
```

The problems:

- **51 do-files** across 4 subdirectories
- Master file from **June 2021** — 3 years stale
- Date-in-filename: `_07sep`, `_02July1130am`
- Author-in-filename: `_eb`, `_e`
- Hardcoded paths — only runs on my machine
- No documentation, no version control

THE PROMPT

What I Asked Claude to Do

```
Create a clean, replicable version of my do-files that generate the analysis in my
Overleaf paper.  I'll give you the file paths.

Archive all old files -- don't delete anything.

The goal is a replication package.  No results should change.

Write a comparison script so I can diff the new output against a "gold standard"
version from the current code.
```

→ One prompt. AI reads 51 files, proposes a new structure, archives the old ones, and builds a test harness.

Code Audit: After

Dimension	Before	After
Do-files	51	**8**
Master file	2021	Current
Config system	None	Template
Version control	None	Git
Documentation	None	Full
"Which version?"	Constant	**Zero**

86 old variants preserved in Archive/ — nothing deleted, fully reversible.

```
bd-remote-ed/
code/00_master_repo.do
code/07_Dofiles_academic/
_globals.do, 02_datagen.do,
03_vargen.do (THE one), ...
Archive/ (86 old)
config/config_local_template.do
docs/   justfile   README.md
```

Time: ∼2 Claude Code sessions (a few hours). **By hand:** days, minimum.

The Bug: A Copy-Paste Typo Hiding in Plain Sight

The `$flags` variable list

`$flags` = 17 missing-value indicators used as controls in **every regression**. Defined independently in 5 scripts.

3 of 5 scripts (wrong):

```
... _f_c_engaged _f_c_health
    _c_child_work
```

Missing the `_f_` prefix

→

Correct:

```
... _f_c_engaged _f_c_health
    _f_c_child_work
```

Result: `_c_child_work` appeared *twice* in regressions. The actual missing indicator was left out. Survived 10+ versions across 2+ years. No error, no crash — just quietly wrong. **How AI found it:** ran `/code-review` on the new code — it cross-referenced every `global flags` definition across all files.

Application 2

Paper Audit with `/econ-audit`

AI vs. 11 Real Referee Reports

Paper Audit: What Is `/econ-audit`?

Adversarial Review Skill

A reusable skill that reads a paper and acts as a **skeptical referee**.

- Reads full paper (LaTeX, tables, appendix)
- Checks identification, power, measurement
- Produces structured audit report
- Takes ∼60 seconds

```
$ npx skills add thinkingwithagents/skills \
-skill econ-audit
```

What it checks:

1. **Identification:** Is the causal claim valid?
2. **Statistics:** MHT, power, attrition
3. **Measurement:** Are variables measured well?
4. **Methodology:** Design consistency
5. **Presentation:** Framing, length

LIVE DEMO

Running /econ-audit on the Paper

```
$ claude
/econ-audit main.tex
```

Reads the full paper + tables + appendix.

Produces a structured audit in ∼60 seconds.

Switching to terminal. . .

AI Audit vs. Real Referees: 7 of 10 in ∼60 Seconds

Referee Criticism	AI?	Notes
High attrition & differential attrition	✔	Flagged as Critical; suggested Lee bounds
Results don't tell a coherent story	◐	Caught ID problem, missed internal contradictions
Theoretical model not useful	✔	Flagged — but recommended *including* it
Short intervention duration (4–8 weeks)	✘	Requires field knowledge
Noisy learning measures (8-item phone test)	✔	Strong match; flagged IRT minimums
Paper too long and hard to follow	✔	Suggested cutting 20–30%
Multiple hypothesis testing issues	✔	Aggressive; added winner's curse framing
Limited external validity / COVID fatigue	◐	Partial; missed publication landscape
Deviations from pre-analysis plan	✔	Caught the spirit, not the specifics
Complex randomization hard to interpret	✔	Strong match

5 full matches, 2 partial, 1 opposite recommendation, 2 misses.
11 referee reports across 8 journals over 4 years — vs. 60 seconds of AI.

What the AI Missed — And Why It Matters

✕ The Misses

- **Short duration:** 4–8 weeks is unusually short for an education RCT. Requires knowing field norms.
- **Internal contradictions:** Data arm increased tutoring *more* than info arm, but only info improved learning. Referees caught this; AI didn't.
- **COVID fatigue:** "We've seen too many of these papers." AI can't know the publication landscape.

The opposite advice:

The AI recommended *including* the conceptual model. Every referee wanted it **cut**.

Sometimes AI gives exactly the wrong recommendation because it lacks field conventions.

Pattern:

AI catches **statistical/econometric** issues reliably.

AI misses **field knowledge**, **within-paper logic checks**, and **publication strategy**.

Takeaway: A Very Good Pre-Submission Check

What It Replaces

- The first 80% of what a careful internal seminar discussant would say
- Catches the systematic stuff: identification, power, MHT, attrition
- In one minute, not one month

What it doesn't replace:

- Field-specific intuition
- "Does this make sense given what we know about Bangladesh?"
- Publication strategy advice
- Colleagues who know your work

For the code audit: it found bugs I didn't find in two years. That alone justified the time.

For the paper audit: run it before you send a paper out. It's free. It catches the embarrassing stuff.

Application 3

"Do Lee Bounds" — From Referee Comment to Results

Acting on the Audit

The Referee Said "Do Lee Bounds"

From the AI audit (and real referees):

"High attrition & differential attrition. **Flagged as Critical.** Suggested Lee bounds."

This is the single most common referee request on this paper.

The question: Can AI just. . . do it?

Write the Stata code for Lee (2009) bounds on the app treatment arm vs. control?

Lee Bounds

- Tightens treatment effect estimates under **differential attrition**
- Trims the "excess" group to equalize response rates
- Gives upper and lower bounds on the true effect
- Standard in development / labor RCTs

LIVE DEMO

Running Lee Bounds

```
$ claude

I need Lee (2009) bounds for the app treatment arm vs.
control.  The outcome is the learning index.  Use the endline
survey flag to define attrition.  Write a Stata do-file, run
it, and show me the upper and lower bounds with confidence
intervals.
```

→ From referee comment to results in one prompt.

What Just Happened

AI Wrote the Code

- Identified the right variables
- Computed attrition rates by arm
- Implemented the trimming procedure
- Produced bounds with CIs
- Formatted a results table

What you still check:

- Is the attrition variable right?
- Is it trimming from the correct tail?
- Do the bounds make sense given your priors?
- Does the code match Lee (2009)?

Time: minutes, not hours.

The audit flagged it. The AI wrote the code. **You check the output.** Same principle, end to end.

Erkmen: Teaching & Applications

Lecture Builder, Research Brainstorm & Web Scraping

Application 3

Lecture Builder Skill — From Papers to a Lecture

Where We Left Off — The Skill Spec

```
/skill-creator

Create a skill called "lecture-builder" that converts academic documents into lecture materials.  The skill should:

1.  Read documents in their entirety (PDF, DOCX, web pages) -- break reading into parts to avoid excessive token

usage.  Include a Python parsing script.

2.  Extract:  key arguments, empirical findings, methodological details, discussion points, potential exam questions.

3.  Generate two outputs:

a) Structured lecture notes (markdown) with section headers, key takeaways, discussion prompts

b) A polished slide deck (draw on \academic-beamer-deck or \pptx) with clean visuals, one idea per slide

4.  Adapt tone for undergraduate vs. graduate audiences (configurable).

Save SKILL.md + supporting files to .claude/skills/lecture-builder/.
```

Last week we wrote the spec. ***Today we actually build the skill — then run it.***

Building the Skill: `/skill-creator` Walkthrough

Building the Skill: `/skill-creator` Walkthrough

`/skill-creator` **interviews you**. Be ready to answer:
name, audience, inputs, outputs, edge cases, where to save.

Anatomy of `lecture-builder`

`.claude/skills/lecture-builder/`

- `SKILL.md` playbook
- `parse_document.py` PDF/DOCX reader
- `slide_template.tex` UVM Beamer
- `notes_template.md` notes scaffold

What `SKILL.md` enforces

- Chunked reading → token control
- Mandatory citations from source PDFs
- Undergrad vs. grad tone toggle
- Dual output: notes + slides
- Reuses UVM color palette

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Skills with Python scripts > pure-markdown skills when tokens matter. The script reads; the LLM synthesizes.

The Input Folder

./inputs/

gov_thinktank_ice_arrests.pdf
policy report

ice_enforcement_report.pdf
government brief

aslim_paper.pdf
academic working paper

Why this mix?

- Heterogeneous sources: **policy** + **government** + **academic**
- Forces the skill to harmonize tone, terminology, and evidence quality
- Mirrors how you'd actually build a lecture

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Why this mix?

- Heterogeneous sources: **policy** + **government** + **academic**
- Forces the skill to harmonize tone, terminology, and evidence quality
- Mirrors how you'd actually build a lecture

Topic: The labor and health consequences of ICE enforcement. Connects directly to ongoing research at UVM.

The Execution Prompt

```
/lecture-builder

I have a folder ./inputs/ with:  gov_thinktank_ice_arrests.pdf,
ice_enforcement_report.pdf, aslim_paper.pdf.

Build a 50-minute undergraduate lecture on the labor and health
consequences of ICE enforcement.  Audience:  junior/senior econ majors
who have taken intermediate micro and one applied econometrics class.

Outputs to ./outputs/:
  1) lecture_notes.md -- sectioned notes with 5 key takeaways,
     4 discussion prompts, 3 candidate exam questions.
  2) lecture_slides.tex -- Beamer deck (~20 slides, 16:9).  Use
     a clean academic palette -- default UVM (deep green / gold) or ask
     me.  In-text citations to all three source PDFs.

Be conservative with tokens:  parse each PDF in chunks via
parse_document.py, summarize per-section, then synthesize.
```

Audience spec + output spec + token discipline. Skills work better when the prompt is concrete.

Expected Output (Live Demo Running)

▶ Live demo running in the background — final files appear by the end of this section

lecture_notes.md

```
# 1. The setting
Recent expansions of interior immigration
enforcement...
# 2. Identification strategy
287(g) rollout as a staggered DiD...
# 3. Findings
Aslim et al. document...
# Discussion prompts
1. Why might the effect on infant
   health attenuate over time?
# Exam questions
1. Sketch the parallel-trends test...
```

synthesizes →

lecture_slides.tex

```
% 20 slides, 16:9
% UVM palette
\begin{frame}{Setting}
  ICE arrests timeline
\end{frame}
\begin{frame}{Methods}
  staggered DiD
\end{frame}
\begin{frame}{Results}
  main effect
\end{frame}
```

What Just Happened (Under the Hood)

Read in chunks

parse_document.py

What Just Happened (Under the Hood)

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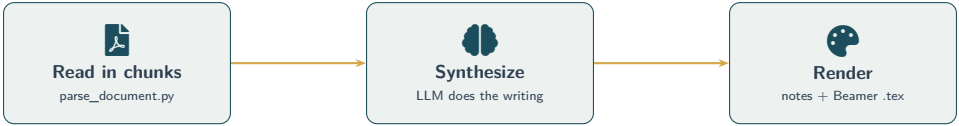
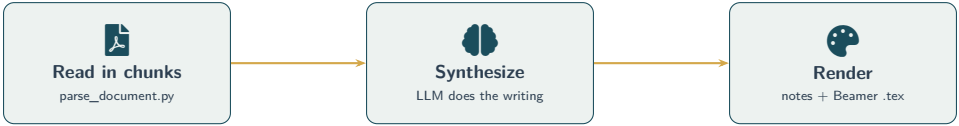


What Just Happened (Under the Hood)



This skill is reusable. Swap the input folder, swap the topic, swap the audience — get a new lecture in minutes. Build once, deploy forever.

Application 4

`/research-brainstorm` — Stress-Test a Research Idea

What the Research Brief Contains

Brief sections

- **The question** — structured: unit, population, treatment, outcome, comparison, setting
- **Closest related work** — 3–6 cites with sharp delta
- **Contribution type** — setting / identification / reframing / measurement / mechanism / policy
- **The strongest objection** — one paragraph, unvarnished
- **Alternative framings** — 2–3 pivots
- **Feasibility** — data, identification, power, timeline, fatal risks
- **Next concrete step**

Saved to:

`research_briefs/YYYY-MM-DD-slug.md`

Markdown so it lives next to your other notes. Re-openable, editable, citeable in your own write-ups.

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The conversation is great.
The artifact is what you keep.

Installing /research-brainstorm

Full economist pack

```
npx skills add
  thinkingwithagents/skills
```

Installs the full pack in one go. Today we use: `research-brainstorm`, `find-data`, `academic-beamer-deck`.

One skill at a time

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Mix-and-match per project. Lighter footprint when you only need one.

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Same `npx skills` mechanism works for any GitHub-hosted skill repo. **Build your own pack and share it with your coauthors.**

Live Demo: Hand Me a Question

Audience:

hand me a half-formed research question.

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hand me a half-formed research question.

An effect
you suspect
but can't yet measure

A dataset
you have access to
but haven't used

A policy
you want
to evaluate

Live: invoke `/research-brainstorm`, ride through the phases, walk away with a brief.

Application 5

Web Scraper — From `/find-data` to a Live Visualization

`/find-data` Picks NBER Working Papers

`/find-data` in one line

We hand it our constraints (applied-micro audience, 24+ months, scrape-able, time-series + cross-section). It scans repositories and surfaces a source.

Today's pick: NBER working papers.

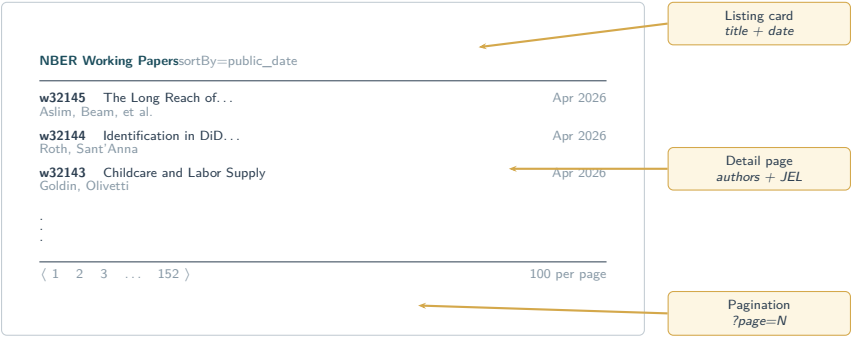
Why NBER: every economist cares; clean HTML; rich JEL + affiliation cross-section; ~1,500 papers / year.

URL:
`nber.org/papers?page=N&perPage=100&sortBy=public_date`

Extract: `nber_number, title, authors, public_date, primary_jel_code, affiliations.`

Goal: novel data on the screen in under 10 minutes. Skip the buildup — go straight to scraping.

Inspect Before You Prompt



View-source first. The agent writes better scrapers when you've already named the selectors.

The Scraping Prompt

```
Scrape NBER working papers for a teaching demo.  Build a self-contained

Python script in ./scrape_nber/ using uv:

1. Fetch the most recent ~1,000 papers from

   nber.org/papers?page=N&perPage=100&sortBy=public_date

   (paginate until 1,000 or cutoff 2024-01-01).

2. Extract:  nber_number, title, authors, public_date, primary

   JEL code, affiliations (from detail page if needed).

3. Save to ./scrape_nber/data/nber_papers.csv.

4. Generate three PNGs to ./scrape_nber/figures/:

   - papers_per_month.png   bars, last 24 months, UVM colors

   - top_jel_codes.png   top 15 primary JEL codes

   - top_institutions.png   top 10 affiliations

5. Print 5-line stdout summary.

Politeness:  real-browser User-Agent; sleep 0.5s between fetches;

cache HTML to ./scrape_nber/cache/ so re-runs are free.
```

What Claude Does While We Keep Talking

Fetching pages...

Parsing...

Plotting...

```
$ uv run scrape_nber/main.py
[INFO] fetching page 1/10 ...ok (98 papers)
[INFO] fetching page 2/10 ...ok (100 papers)
[INFO] ...
[INFO] parsing JEL codes ...
[INFO] saved data/nber_papers.csv
[INFO] writing figures/...

== Summary ==
n_papers:  1,000
date range:  2024-04 to 2026-04
top JEL: J (Labor)
top affil:  NBER, Harvard, MIT
```

The terminal is a TV. Let it run while you talk through what's coming.

What Claude Does While We Keep Talking

✔ Fetched 10 / 10 pages

Parsing 1,000 records. . .

Plotting. . .

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◌ **Plotting 3 figures. . .**

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What Claude Does While We Keep Talking

- ✔ Fetched 10 / 10 pages
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- ✔ Wrote 3 PNGs

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What to Try First

Start Small, Check Each Step

What to Try First

#	Activity	Time	What You Get
1	Create a voice file	15 min	Your tone in every AI draft — reuse forever
2	Data pipeline (describe → propose → code)	30 min	Orientation on any dataset
3	Install `/econ-audit`, run on a paper	10 min	Pre-submission check for free
4	Try inbox triage with Gmail integration	30 min	Sorted, triaged, draft replies

All four use the same principle: **AI does the mechanical parts, you check each step.** Start with whichever one matches a task you already have.

What We Built Today

1. Code audit: **51 files to 8**, found a bug hiding for 2+ years

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Six workflows. Six artifacts on disk. **All reusable.**

Resources

Links

- **Bootcamp website:**
  `thinkingwithagents.github.io`
- **Skills library:**
  `skills.sh`
- **Econ skills:**
  `github.com/thinkingwithagents/skills`
- **Claude Code docs:**
  `docs.anthropic.com/en/docs/claude-code`

Install Commands

Step 1: Install Node.js
`nodejs.org` → click LTS

Step 2: Install Claude Code
```
npm install -g @anthropic-ai/claude-code
```

Step 3: Install econ skills
```
npx skills add thinkingwithagents/skills
```

Step 4: Launch
```
claude
```

Thinking With Agents

Online sessions coming this summer

Workshops, deep-dives, and Q&A on agentic AI for academic research — built by economists, for economists and academics alike.

Stay tuned for dates and registration:

`thinkingwithagents.github.io`